Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	27 October 2026
Team ID	
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Tableau Calculated Fields	Enables calculations, conditional logic, and analysis of patient health indicators without external programming.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	CSV /Microsoft excel	Stores patient demographics, BMI, blood pressure, glucose, cholesterol, disease conditions, and lifestyle factors in a structured format.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Data Connections	Allows direct integration of healthcare datasets into Tableau for further processing and visualization.

4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Calculated Fields,Filters,Aggregations	Supports age-group classification, BMI categorization, disease prevalence calculations, and identification of health risk patterns.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Public	Helps visualize disease trends, compare patient groups, analyze lifestyle factors, and identify potential healthcare risks.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public Dashboard and story	Provides an accessible, interactive interface for exploring patient health trends and supporting informed healthcare decisions.